Research on multiplexed colour QR code with direct readability

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With the rapid popularization of QR code technology, colour QR code is an important direction of its development, making it of great value to research a colour QR code that can be read directly while improving the data capacity. This letter proposes a novel composite colour QR code by combining colour coding and multiplexing technology. Based on the actual demands of high-capacity, this code presents the public identification information in the form of the plain code and conceals the secret information (e.g., authentication information such as traceability, blockchain) as the hidden code. The reading conditions of colour QR codes are discussed, a colour coding model is established and the corresponding algorithm to optimize its design is given. Through code scanning tests, it is verified that this kind of code has significant value in the field of anti-counterfeiting, commodity marketing etc.

Introduction: QR code is widely used in various fields for its high capacity and easy reading. To meet different practical requirements, its storage capacity and anti-counterfeiting are currently urgent issues to be addressed. The founder of QR code pointed out that colour QR code [1] is a paramount direction for its future development. The development of colour QR codes is severely constrained during application owing to its dependence on special software and hardware systems. Colour QR code has become a hot topic in the research of QR code because of the functional characteristics carried by their colours, most of which focus on colour QR code with high-capacity [2].

It is the most direct and cheapest method to improve the information storage capacity of QR codes by introducing different colours. Colour coding technology [3] and multiplexing technology [4] are the most commonly used technical means of colour QR codes. Nutchanad Taveerad et al. [5] realized the raise of data density by encoding and decoding information with custom colours. Pillai, et al. [6, 7] increased the data capacity of QR code and ensuring its robustness based on colour coding technology. Galiyawala, et al. [8, 9] applied multiplexing technology [10] to compound multiple QR codes into a colour QR code, whose capacity is 24 times that of black-and-white QR code of the same size.

Although the related technology of colour QR codes solves the capacity problem of monochrome QR code, it also unavoidably brings new matters: on the one hand, the multiplexed QR codes need to be decoded with the help of special reading software, which greatly hinders its integration and application; on the other hand, the additional colours are affected by environmental conditions, resulting in lower accuracy of decoding, which seriously restricts its technology landing. This letter provides a new type of multiplexed colour QR code (abbreviated as "MC2D code"), its main contributions are as follows:

- We combine colour coding technology and multiplexing technology and improve the capacity while increasing the anti-counterfeiting.
- We compound a variety of different types of QR codes into a colour QR code, and plain and hidden codes have good compatibility.
- MC2D code can be read directly using ordinary scanning equipment or software (such as WeChat and Alipay).
- MC2D code is processed by optical filtering to realize the effective distinction between plain and hidden codes, which is suitable for any scene requiring multiple QR codes.

Design of MC2D code: The decoding core of QR codes is to divide the module state based on grayscale. Through the normalization of the local grayscale of a single module, it is judged whether the current module is the dark or light state, which is converted into an array represented 0 and 1 for decoding. The existing colour QR code applies hue to identify different colours for custom encoding and decoding, whose colours are only used as the carrier of information. When this type of colour QR code is decoded, it is prone to appear grayscale confusion in a large range, resulting in decoding failure, and the information it carries needs to be obtained by special software or equipment.

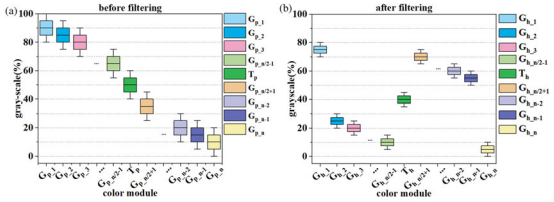


Fig. 1 *Colour coding model of MC2D code*

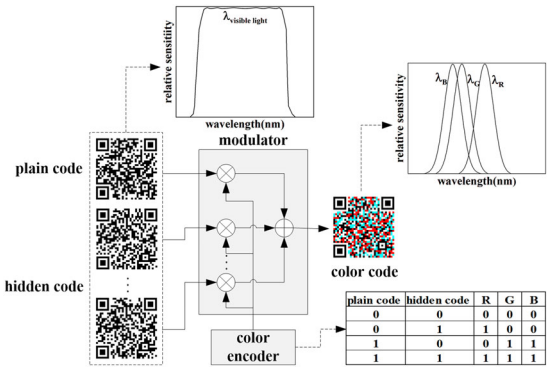


Fig. 2 *Colour modulation of MC2D code (QR code)*

Aiming at the problem of colour QR code reading, we provide a colour coding model based on grayscale and wavelength, as shown in Figure 1. Assuming that N QR codes are multiplexed, at least 2^N colour modules with different wavelengths should be used, which can represent 2^N grayscales and 2 steady states.

The n colour modules before filtering are based on grayscale to characterize the plain code, denoted as G_{p_i}, $(i = 1, 2, \ldots, n)$, $G_{p_n} \le G_{p_n-1} \le \ldots \le G_{p_n/2+1} \le T_p \le G_{p_n/2} \le \ldots \le G_{p_2} \le G_{p_1}$; each module has a bistable grayscale jump (mutual jump between dark and light states) after filtering due to the different wavelengths, and the hidden code is represented by the grayscale after the jump, recorded as G_{h_i}, $(i = 1, 2, \ldots, n)$, $G_{h_n} \le G_{h_2} \le \ldots \le G_{h_n/2} \le T_h \le G_{h_n-1} \le \ldots \le G_{h_n/2+1} \le G_{h_1}$. To ensure that the generated MC2D code can detect different effective QR codes, it is necessary to improve the anti-interference tolerance between modules as much as possible, and strictly avoid the misjudgement of grayscale states. Where T_p and T_h are grayscale judgment thresholds of plain and hidden codes respectively, which are solved by (1). Grayscale discrimination is performed for each module according to the judgment thresholds.

$$\begin{cases} T_p = (G_{p_\max} + G_{p_\min})/2 \\ T_h = (G_{h_\max} + G_{h_\min})/2 \end{cases} \tag{1}$$

where G_{p_max}, G_{p_min}, G_{h_max}, G_{h_min} are the maximum and minimum grayscale of n modules before and after filtering.

Plain and hidden codes are standard black and white QR codes, the plain code is used as the carrier image, and one or more hidden codes are selected as the modulation information through the modulator to encode colour and grayscale, so as to generate the multiplexed colour QR code. The colour modulation process is shown in Figure 2.

Based on RGB, CMYK, or Lab colour space, each base colour and its mixed colour can produce colours in any grayscale (taking RGB colour space as an example). Plain and hidden codes represented by "0" and "1" are used as the input of the colour encoder. The R, G, B are used as the output, with the value of [0,1] indicating the ratio of their use. According to this information modulation method, traversal modulation is performed on plain and hidden codes.

We use three kinds of optical filters to investigate colour QR codes, denoted as F_r, F_g and F_b, with wavelengths λ_r, λ_g and λ_b. The wavelengths of the three primary colours are recorded as λ_R, λ_G and λ_B. Optical filters are narrow-band filters, only part of wavelengths of colours can pass (indicated by "1"), while colours not in their wavelength range are filtered out (indicated by "0"), which we call the colour selection process. The colour gating results of RGB by the filter are shown in Table 1.

Table 1. Colour gating results

Optical filter	Wavelength range (nm)	Relationship between the wavelength			R	G	B
F_r	$\lambda_r \in [620, 650]$	$\lambda_R \in \lambda_r$	$\lambda_G < \lambda_r$	$\lambda_B < \lambda_r$	1	0	0
F_g	$\lambda_g \in [520, 550]$	$\lambda_R > \lambda_g$	$\lambda_G \in \lambda_g$	$\lambda_B < \lambda_g$	0	1	0
F_b	$\lambda_b \in [480, 510]$	$\lambda_R > \lambda_b$	$\lambda_G > \lambda_b$	$\lambda_B \in \lambda_b$	0	0	1

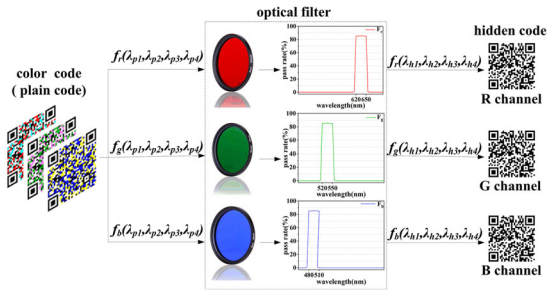


Fig. 3 *Plain and hidden codes representation of MC2D code*

We use the wavelength relationship: $\lambda_B < \lambda_G < \lambda_R$ to select the proportions and combinations of RGB for different optical filters. taking the composite of two QR codes as an example, the colour conversion process based on optical filters F_r, F_g and F_b is shown in (2), (3) and (4):

$$\begin{bmatrix} 0 & 0 & 0 \\ k_{r1} & 0 & 0 \\ 0 & k_{r2} & k_{r3} \\ k_{r4} & k_{r5} & k_{r6} \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 0 \\ k_{r1}R \\ k_{r2}G + k_{r3}B \\ k_{r4}R + k_{r5}G + k_{r6}B \end{bmatrix} \tag{2}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & k_{g1} & 0 \\ k_{g2} & 0 & k_{g3} \\ k_{g4} & k_{g5} & k_{g6} \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 0 \\ k_{g1}G \\ k_{g2}R + k_{g3}B \\ k_{g4}R + k_{g5}G + k_{g6}B \end{bmatrix} \tag{3}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & k_{b1} \\ k_{b2} & k_{b3} & 0 \\ k_{b4} & k_{b5} & k_{b6} \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 0 \\ k_{b1}B \\ k_{b2}R + k_{b3}G \\ k_{b4}R + k_{b5}G + k_{b6}B \end{bmatrix} \tag{4}$$

where the first matrix on the left side of the equation is the coefficient matrix, k_{ri}, k_{gi}, k_{gi} (i = 1, 2, …, 6) represent the proportion of RGB, the second matrix is three colour components, and the right side of the equation is the colour combination used to generate colour QR code.

MC2D code uses the gating characteristics of optical filters to make it behave as the plain code before filtering, and the wavelength of each module is marked as $f_r(\lambda_{p1},\lambda_{p2},\lambda_{p3},\lambda_{p4})$, $f_g(\lambda_{p1},\lambda_{p2},\lambda_{p3},\lambda_{p4})$, $f_b(\lambda_{p1},\lambda_{p2},\lambda_{p3},\lambda_{p4})$, the light and dark states of some modules are switched, and output the hidden code composed of modules with wavelengths $f_r(\lambda_{h1},\lambda_{h2},\lambda_{h3},\lambda_{h4})$, $f_g(\lambda_{h1},\lambda_{h2},\lambda_{h3},\lambda_{h4})$, $f_b(\lambda_{h1},\lambda_{h2},\lambda_{h3},\lambda_{h4})$. The representation of plain and hidden codes is shown in Figure 3.

Algorithm for MC2D code: MC2D code is the result of the multiplexing of plain and hidden codes through information modulation. QR codes used are both binary images. By judging both the grayscale state in each pixel to choose custom colour matching, the algorithm steps are shown in algorithm 1.

Optimal design of MC2D code: To design the optimal MC2D code, the grayscale with different combinations of colour modules are discussed, and the minimum grayscale difference that can be bistable resolved by the reading equipment is found. In the case of multiplexing two QR codes, at least four colour modules are required. Since $G_{p_4} \leq G_{p_3} < T_p < G_{p_2} \leq G_{p_1}$ and $G_{h_4} \leq G_{h_2} < T_h < G_{h_3} \leq G_{h_1}$, the readability of the plain code mainly depends on whether G_{p_2} and G_{p_3} can be correctly judged by T_p, and the judgment result of G_{h_2} and G_{h_3} by T_h determines whether the hidden code can be read. Grayscale test

Algorithm 1 Generation of MC2D code

1: **Input**: Plain and hidden code images: img_p, img_h
2: Get the size (H × W) of img_p
3: Define four sets of colour ratios [R_i, G_i, B_i], i∈[1, 4]
4: **for** i = 1:H
  for j = 1: W
5: Select the RGB ratio according to the grayscale of each pixel img_p and img_h.
  img_c(i, j)=[R_i,G_i,B_i], i∈[1,4]
6: **end; end;**
7: **Output**: the generated colour QR code image: img_c

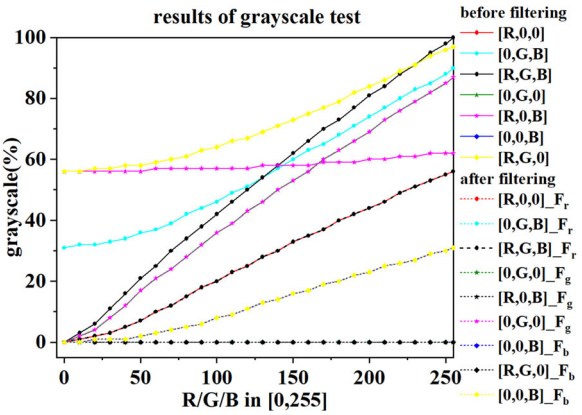


Fig. 4 *Grayscale test results of colour modules*

Table 2. Readable grayscale redundancy of MC2D code

Code	Condition A (%)	Condition B (%)	Results
Plain code	$15 \leq G_{p_2} - T_p < 20$	$15 \leq T_p - G_{p_3} < 20$	Slow
	$20 \leq G_{p_2} - T_p$	$20 \leq T_p - G_{p_3}$	Normal
Hidden code	$7 \leq T_h - G_{h_2} < 11$	$7 \leq G_{h_3} - T_h < 11$	Slow
	$11 \leq T_h - G_{h_2}$	$11 \leq G_{h_3} - T_h$	Normal

is conducted for colour combinations in (2), (3) and (4), as shown in Figure 4.

Based on the grayscale test results in Figure 4, the colour ratios that satisfy the colour coding model are selected and MC2D code (plain code) is generated according to the algorithm steps in algorithm 1. Using the image processing software (such as photoshop) to simulate the ideal narrowband filtering effect to obtain the hidden code. We tested MC2D code (electronic images on a screen with a display resolution of 1920 × 1080) with a common scanning device for reading. If plain and hidden codes cannot be read simultaneously, iteratively correct the relevant parameters to increase the grayscale difference between bistable states until both can be decoded, and the grayscale and proportion of each module at this time be recorded.

Optimization design is simply to find the range of grayscale difference between G_{p_2} and G_{p_3}, G_{h_2} and G_{h_3} that can be distinguished by T_p and T_h. According to the above readable MC2D code, the control variable method is used to keep three colour modules unchanged, and the other colour module is studied. For the plain code: G_{p_1}, G_{p_2}/G_{p_3} and G_{p_4} remain unchanged, applying a positive and negative grayscale offset to the module G_{p_3}/G_{p_2} with a fixed step. The two critical values between the normal reading, slow reading and inability to read are recorded, that is, the two intervals of normal and slow reading, and the same for the hidden code. The readable grayscale redundancy of MC2D code is shown in Table 2.

When the grayscale of MC2D code before and after filtering meets the two conditions shown in Table 2 to satisfy the reading results, otherwise it cannot be read (both plain and hidden codes can be read, said MC2D code can be read directly).

Results and evaluation: It can be seen from Figure 4 that green is a "bright colour", which represents a larger grayscale compared with the same proportions of red and blue, and has a poor presentation ability for dark colour, making it difficult to achieve a grayscale jump between

Table 3. Directly readable MC2D code

Code	Plain code_F_r	Hidden code_F_r	Plain code_F_b	Hidden code_F_b
PDF 417				
QR code				

Table 4. Reading stability evaluation of MC2D code

Plain/ Hidden code	Resolution (dpi)		Size (height)		Reading angle (°)			
	1366 × 768	1920 × 1080	Size <1 cm	Size ≥1 cm	0	90	180	270
Huawei	√/√	√/√	×/√	√/√	√/√	√/√	√/√	√/√
Xiaomi	√/√	√/√	×/√	√/√	√/√	√/√	√/√	√/√
Vivo	√/√	√/√	×/√	√/√	√/√	√/√	√/√	√/√
Samsung	√/√	√/√	×/√	√/√	√/√	√/√	√/√	√/√
Apple	√/√	√/√	×/√	√/√	√/√	√/√	√/√	√/√

Table 5. Performance comparison of MC2D code with some of the state-of-the-art results

Readability	HIQ [7]	CQR [11]	HD2DC [12]	MC2D (ours)
Common decoder	×	×	×	√
Optical filter	×	×	×	√
Coloured light	×	×	×	√

bistability. Therefore, the experimental results of MC2D code based on F_r and F_b filters are shown in Table 3.

By scanning QR codes in Table 3, it is found that these codes can directly read the loaded multi-code information with ordinary scanning equipment, and plain and hidden codes have great compatibility.

The stability of QR code reading is affected by many external factors. For electronic images, it is mainly displayed resolution of the screen, size of code and reading angle. These effects are multiplied for colour QR code. We investigated these three factors on the reading stability of MC2D code: fixing the size and reading angle, and changing the display resolution of the screen to get the effect of resolution on the reading stability. Similarly, study the other two factors. We use five different models of mobile phones to perform plain and hidden codes scanning tests on the MC2D code ("√" means stable, and "×" means unstable). The test results are shown in Table 4. In addition, we also compared the proposed MC2D code with the recent research results of colour QR codes, mainly for its direct readability in different environments ("√" means decodable, "×" means not decodable), and the results are shown in Table 5.

As can be seen from Tables 4 and 5, the reading result of the MC2D code we designed on different models of mobile phones are the same. Under different screen resolutions and reading angles, the reading of plain and hidden codes can be read stably. When the size is less than 1 cm (with height as the standard), the reading of the plain code is affected to a certain extent, but in practical applications, its size is generally not too small, so it will not affect the use, which proves that MC2D code has strong stability in reading. In terms of readability, compared with the more typical colour QR codes such as HIQ, CQR, HD2DC etc., MC2D code has the advantage of being directly readable under different conditions and better applicability.

Conclusion: In this letter, we provide a novel colour QR code with multi-code composited, which combines colour coding and multiplexing technology to construct colour coding model and algorithm to generate this type of QR code, and we find its readable grayscale redundancy through optimal design. This code solves the compatibility problem of multi-code integration, breaking through the technical matter that existing colour QR codes cannot be read directly. Users can obtain the information of plain and hidden codes without special equipment. Experiments have proved that this code has the advantages of innovation and multi-functionality, which can meet any practical demands requiring plain and hidden information, and has great development potential in application and promotion. Future research can focus on halftone colour QR codes and improve the applicability of colour QR codes to accelerate its technology implementation.

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